

ARCHITECT REVIEW

SALDEV-1413 | Stepped-up Pricing ARR Flows

Flywire Partial sandbox | Read-only implementation and promotion-readiness assessment

Environment	FlywirePartial Org 00DhG0000000jOX Sandbox
Review date	26 August 2026
Evidence	Active-state read-back retrieved metadata Apex test SOQL collision checks
Decision	Changes Requested - not ready for promotion
Boundary	No code edits, deployment, activation, data changes, or Production changes

ARCHITECT DECISION

Do not promote yet. The implementation is bulk-aware and metadata-driven, but its aggregation key merges distinct lines, seven active flows conflict with the latest story clarification, and rollup failures can leave unsafe prorate divisors.

Acceptance outcome

#	Scope	Result	Evidence
1	Aggregation correctness	Fail	Apex groups only by Quote + Product; sandbox evidence found 20+ groups with multiple distinct line identities.
2	Requirement alignment	Fail	The final story comment says retain Quoted Usage fields, but two mappings and seven active flows average them.
3	Runtime durability	Fail	Partial SaveResults are ignored and three flows divide directly by Total_Prorate_Multiplier__c.
4	Active component scope	Pass	All 12 named target flows are active/latest; the active B2B XB API name is the _1 definition.
5	Automated proof	Partial	Focused Apex tests passed, but handler coverage is 72% and four mappings plus Flow behavior are untested.

Correction and control path

Use a targeted correction: preserve the metadata-driven rollup engine, group by a stable ramp-line identity, align the seven affected flows with the final product-owner decision, make rollup writes atomic, and prove the result with focused Apex and business UAT.

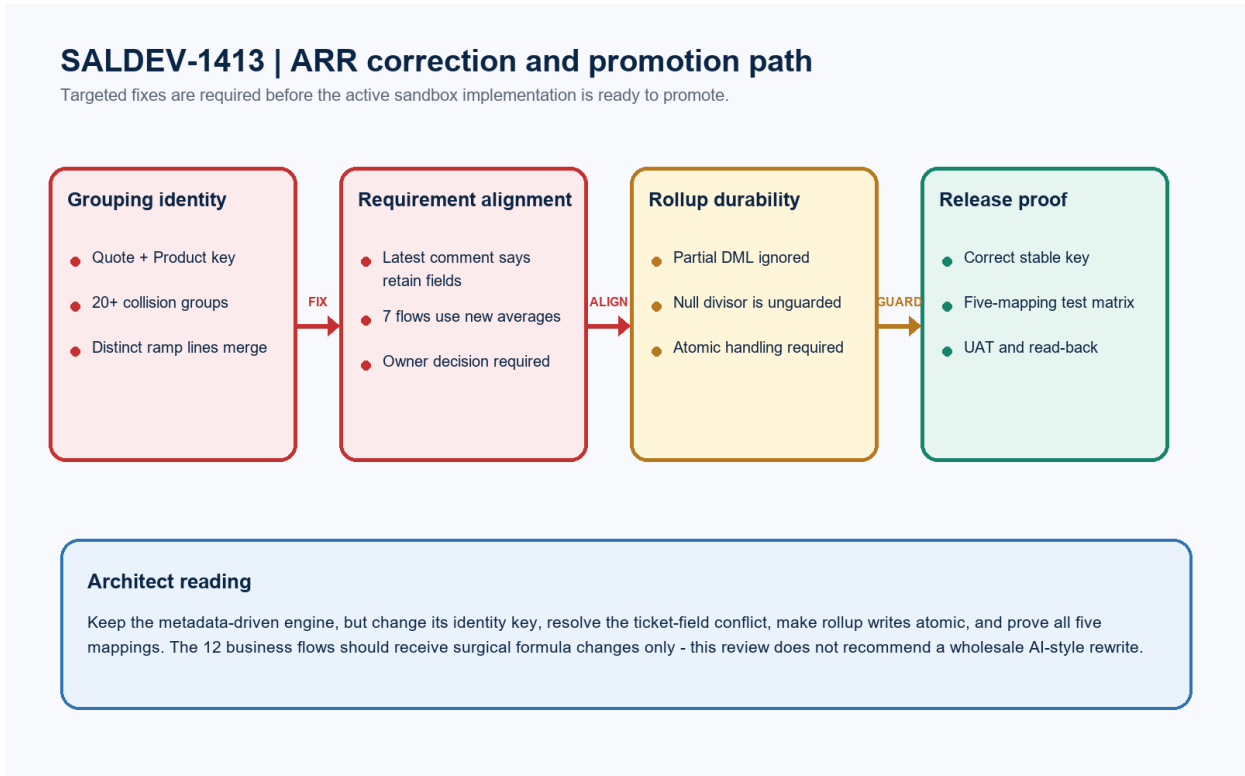


Figure 1. SALDEV-1413 ARR correction and promotion path

DESIGN READING

Red marks release blockers. Amber marks runtime risk. Green is the required promotion proof. Correct the shared engine and formulas surgically; do not rewrite all 12 flows.

Current SALDEV-1413 implementation evidence

Apex and configuration evidence

Evidence area	Current implementation	Acceptance	Architect reading
Group identity	Quote + Product	Fail	The key can merge distinct product occurrences and ramp lines.
Requirement mapping	5 CMDT mappings	Fail	Two usage mappings conflict with the final story comment.
DML handling	update(..., false)	Fail	Unchecked SaveResults allow partial or stale rollups.
Prorate safety	Direct division	Fail	Three active flows lack null/zero divisor guards.
Audit narrative	Healthcare text mismatch	Fail	The ARR formula and explanation use different fields.
Bulk structure	2 queries + 1 DML	Pass	No SOQL or DML occurs inside rollup loops.

Flow and test evidence

Evidence area	Observed state	Acceptance	Architect reading
Usage-field flows	7 active flows	Fail	Seven flows replace fields the final comment says to retain.
Prorate flows	3 active flows	Fail	Three flows divide by an unguarded total.
Collision evidence	20+ groups	Fail	One group had 395 lines across 394 identities.
Test coverage	5 methods; 72%	Fail	Only rate is asserted; no five-mapping identity matrix.

Evidence area	Observed state	Acceptance	Architect reading
Definition state	12/12 active/latest	Pass	Retrieved versions match the target list.

PROMOTION BLOCKER

Functional correctness is not established for duplicate product occurrences, the final usage-field decision, null prorate totals, or all five mappings. These gaps must close before promotion.

Implementation and release boundary

Component type	Focused scope	Required action
Apex rollup service	QuoteLineTriggerHandler.calculateAveragePrice	Group by Opportunity_Line_Id__c or an approved composite identity.
Quote Line trigger	QuoteLineTrigger after-save path	Preserve bulk execution; make recursion reset and error handling deterministic.
Custom metadata	5 Quote Line rollup mappings	Validate fields and operations; retain/remove usage mappings after owner decision.
Usage ARR flows	7 active flows	Apply only the approved usage-field formula decision; preserve unrelated logic.
Prorate ARR flows	Edu Payables; Fixed Price; Travel Payables	Guard null/zero totals and require atomic rollup completion.
Automated tests	Apex plus targeted Flow/UAT scenarios	Cover five mappings, duplicate identities, bulk quotes, delete/update, null/zero, and regression.

Release sequence

Phase	Required evidence
Pre-deployment	Obtain written owner confirmation on Quoted Usage fields; approve the stable line identity; retrieve the current target-org baseline.
Sandbox correction	Make a focused Apex/mapping/formula change only; preserve unrelated Flow behavior and active-state boundaries.
Validation and UAT	Run the expanded Apex matrix, focused Flow tests, check-only validation, stepped/non-stepped business scenarios, and capture job IDs and record-level expected ARR.
Promotion and read-back	Promote only after approval; verify active versions, Apex coverage, rollup fields, ARR outputs, and logs. Roll back the approved package if a critical regression appears.

PRODUCTION GATE

This review covered FlywirePartial only. Production was not inventoried or changed. Re-retrieve current Production metadata and run focused check-only validation before any deployment decision.

UAT and approval gates

Test	Pass evidence
Stepped grouping	Two occurrences of the same product retain separate identities while each ramp sequence receives its own average and total proration.
Non-stepped regression	A non-stepped line produces the same ARR as the current approved baseline with no average-field side effect.
Usage-field decision	Percent and amount scenarios match the written product-owner decision, including the existing divide-by-100 behavior.
Prorate products	Edu Payables, Fixed Price, and Travel Payables handle populated, null, and zero total-prorate conditions without blank ARR or Flow faults.
Bulk and lifecycle	Multi-quote insert, update, delete, start-date shift, and duplicate-product tests pass with expected rollup placement.

Required approval gates

#	Required action	Owner	Evidence needed
1	Confirm usage-field requirement	Product Owner / Architect	Written decision resolving the final comment versus earlier design text.
2	Approve rollup identity	CPQ Architect	Named stable key and examples for repeated products, bundles, and ramp groups.
3	Complete focused correction	Development Lead	Reviewed Apex, CMDT, and Flow diff with no unrelated rewrite.
4	Validate sandbox package	Release Manager / QA	Expanded test results, coverage, check-only job ID, UAT records, and expected values.
5	Promote and read back	DevOps / Business Owner	Deployment receipt, active-version checks, field-level read-back, ARR results, and sign-offs.

Ready-to-paste architect response

STORY COMMENT

Architect review completed for SALDEV-1413 in FlywirePartial. Changes Requested. All 12 target flows are active/latest and focused Apex tests passed, but Quote + Product grouping can merge distinct line identities. The final story comment says retain Quoted Usage Percent and Amount; two mappings and seven active flows average them. Partial rollup failures are ignored, and three flows divide by Total_Prorate_Multiplier__c without a null/zero guard. Confirm the usage-field requirement, approve a stable line key, make writes atomic, protect the divisor, and run the five-mapping stepped/non-stepped matrix before promotion. Retrieval 09ShG000006FuUgUAK; Apex 707hG00000LJDjIQA. No code, metadata, activation, deployment, data, or Production changes were made.